```
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Contents for Theory:
Data Analytics III
1. Implement Simple Naïve Bayes classification algorithm using Python/R on
iris.csv dataset.
2. Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate,
Precision, Recall
on the given dataset.
```

In [1]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix
```

In [2]:
```
df = pd.read_csv("iris.csv")
```

In [4]:
```
df
```

Out[4]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

In [5]:
```
X = df.drop('Species', axis=1)
y = df['Species']
```

In [6]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)
```

In [7]:
```
model = GaussianNB()
model.fit(X_train, y_train)
```
Out[7]:
```
▾ GaussianNB

GaussianNB()
```

In [8]:
```
y_pred = model.predict(X_test)
```

In [10]:
```
import numpy as np
```

In [11]:
```
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)
```
```
Confusion Matrix:
 [[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```

In [12]:
```
TP = np.diag(cm)
FP = np.sum(cm, axis=0) - TP
FN = np.sum(cm, axis=1) - TP
TN = np.sum(cm) - (TP + FP + FN)
```

In [13]:
```
accuracy = np.sum(TP) / np.sum(cm)
```

In [14]:
```
error_rate = 1 - accuracy
```

In [15]:
```
precision = TP / (TP + FP)
```

In [16]:
```
recall = TP / (TP + FN)
```

In [17]:
```
print("\nAccuracy:", accuracy)
print("Error Rate:", error_rate)
print("Precision:", precision)
print("Recall:", recall)
```
```

Accuracy: 1.0
Error Rate: 0.0
Precision: [1. 1. 1.]
Recall: [1. 1. 1.]
```

In [ ]:
```

```